

From ID: 0x92E (Status 1) -> B0 B1 B2 B3 B4 B5 B6 B7

B0: ERPM 1

B1: ERPM 2

B2: ERPM 3

B3: ERPM 4

B4: Current high

B5: Current low

B6: Duty cycle high

B7: Duty cycle low

Example: 00 00 00 00 00 00 01 2B means 0x012B for duty cycle, which is 299 in decimal, divide by 10 is 29.9% duty cycle

Example: 00 00 00 00 01 2A 01 2B means 0x012B for duty cycle, which is 299 in decimal, divide by 10 is 29.9% duty cycle. It means 0x012A for current, which is 298 in decimal, divide by 10 is 29.8A

Example: 00 00 24 C1 01 2A 01 2B means 0x012B for duty cycle, which is 299 in decimal, divide by 10 is 29.9% duty cycle. It means 0x012A for current, which is 298 in decimal, divide by 10 is 29.8A. It means 0x000024C1 for ERPM, which is 9409 in decimal, no multiplier so it is 9409 ERPM

From ID: 0xE2E (Status 2) -> B0 B1 B2 B3 B4 B5 B6 B7

B0: Amp hours used 1

B1: Amp hours used 2

B2: Amp hours used 3

B3: Amp hours used 4

B4: Amp hours charged 1

B5: Amp hours charged 2

B6: Amp hours charged 3

B7: Amp hours charged 4

Example: 00 00 00 A6 00 00 00 03 means 0x000000A6 for amp hours used in milliamp hours, divided by 10. 0x000000A6 is 166 in decimal, divide by 10 is 16.6 mAh. 0x00000003 for amp hours charged in milliamp hours, divided by 10. 0x00000003 is 3 in decimal, divide by 10 is 0.3 mAh.

From ID: 0xF2E (Status 3) -> B0 B1 B2 B3 B4 B5 B6 B7

B0: Watt hours used 1
B1: Watt hours used 2
B2: Watt hours used 3
B3: Watt hours used 4
B4: Watt hours charged 1
B5: Watt hours charged 2
B6: Watt hours charged 3
B7: Watt hours charged 4

Example: 00 00 21 45 00 00 00 A6 means 0x00002145 for watt hours used in watt hours, divided by 10000. 0x00002145 is 8517 in decimal, divide by 10000 is 0.8517 Wh. 0x000000A6 for watt hours charged in watt hours, divided by 10000. 0x000000A6 is 166 in decimal, divide by 10000 is 0.0166 Wh. On the VESC config tool, this data is rounded to nearest two decimal places.

From ID: 0x102E (Status 4) -> B0 B1 B2 B3 B4 B5 B6 B7

B0: IGBT Temp High
B1: IGBT Temp Low
B2: Motor Temp High
B3: Motor Temp Low
B4: Input Current High
B5: Input Current Low
B6: PID Position Now High
B7: PID Position Now Low

Example: 01 45 01 01 00 00 3D 86 means 0x0145 for IGBT temperature, divided by 10. 0x0145 is 325 in decimal divide by 10 is 32.5C. 0x0101 is for Motor temperature divided by 10. 0x0101 is 257 in decimal, divide by 10 is 25.7C. **INSERT CURRENT IN EXAMPLE HERE.** 0x3D86 for PID Position Now is divided by 50. 0x3D86 is 15750 in decimal, divided by 50 is 315 degrees. This relates to the approx rotor position at any given time. At rest with HFI, the default position is 315 degrees.

From ID: 0x1B2E (Status 5) -> B0 B1 B2 B3 B4 B5 B6 B7

B0: Tach 1
B1: Tach 2
B2: Tach 3
B3: Tach 4
B4: DC Voltage High
B5: DC Voltage Low
B6: Not used
B7: Not used

Example: 00 00 6B 78 02 69 00 00 means 0x00006B78 for Tac, which is 27512 in decimal and is not multiplied. 0x02690000 means 0x0269 for DC voltage input, which is 617 in decimal, divide by 10 to get 61.7 Volts input.

From ID: 0x3A2E (Status 6) -> B0 B1 B2 B3 B4 B5 B6 B7

B0: ADC 1 High
B1: ADC 1 Low
B2: ADC 2 High
B3: ADC 2 Low
B4: ADC 3 High
B5: ADC 3 Low
B6: PPM High
B7: PPM Low

Example:

COMMANDS

Message Name	Message Extended ID	Byte 1	Byte 2	Byte 3	Byte 4	Byte 5	Byte 6	Byte 7	Byte 8
Set Duty Cycle	0x000000TA	Duty Cycle * 100000							
Set Current	0x000001TA	Current * 1000							
Set Current Brake	0x000002TA	Brake Current * 1000							
Set RPM	0x000003TA	RPM							
Set Pos(ition?)	0x000004TA	Pos(ition?) * 1000000							
Get Data Long Buff (when instruction 0x04 is sent with message 8)	0x000005TA (TA in this case is the node that asked for the data, which is byte 1 in message 8)	0x00	0x04	mostemp1		mostemp2		mostemp3	
		0x07	mostemp4		mostemp5		mostemp6		temppcb
		0x0E	temp pcb	avg motor current			avg input current		
		0x15	avg input current		duty cycle		rpm		
		0x1C	rpm		voltage		amp hours consumed		
		0x23	amp hours charged				watt hours consumed		
		0x2A	whc		watt hours charged			odometer	
		0x31	odometer		odometer abs			fault code	
Process Buffer (acts as flag for 5)	0x000007TA	SA	0x01	total buffer length		crc			
Ask For Data	0x000008TA	SA	0x00	instruction					
Status Broadcast	0x000009SA	rpm				current		duty cycle	

Set current example:

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unsigned char data[4] = {0x00, 0x00, 0x0F, 0xFF};
CAN.sendMsgBuf(0x00000100, 1, 4, data);
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Here, B1, 2, 3, 4 are read left to right. B1 = 0x00, B2 = 0x00, B3 = 0x0F, B4 = 0xFF

The number is constructed as normal numbers are. 0x00000FFF = 4095

Divide the number by 1000 to get current in amps

Ex. 4095 / 1000 = 4.095A commanded